

MSPM0_K210 Color Recognition

1. K210 and MSPM0 Communication

1.1 Experiment Prerequisites

This tutorial uses the MSPM0G3519 development board. K210 needs to run the program in **K210-AI(MSPM0)** to start the experiment.

MSPM0G3519 *1

K210 vision module *1 (with SD card (with AI model), camera)

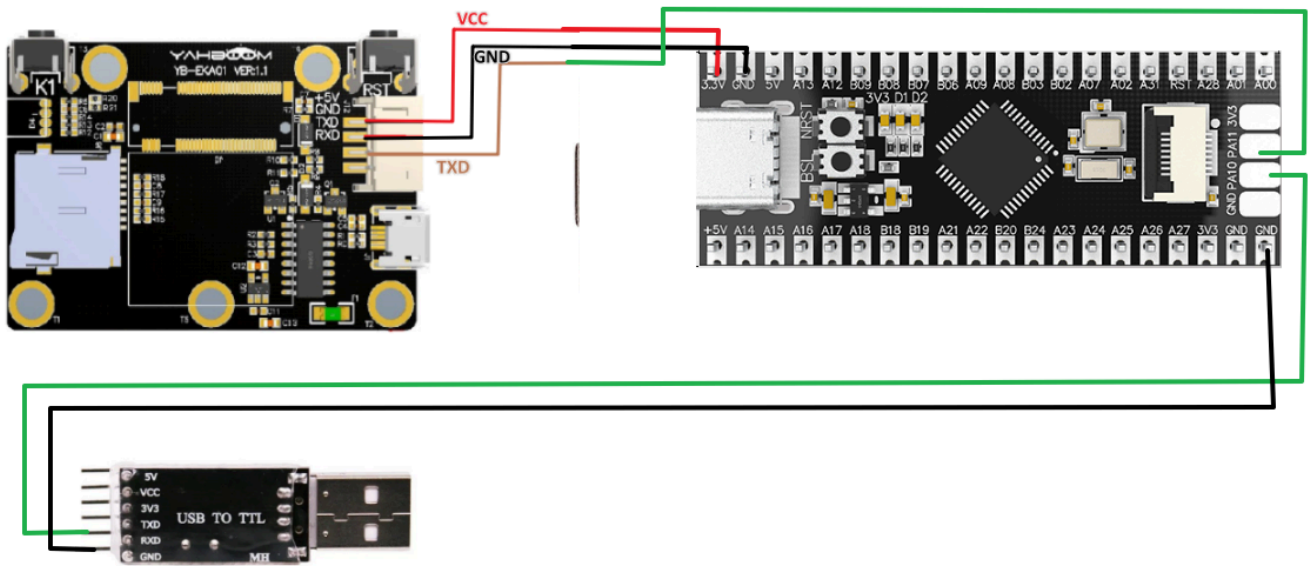
USB to TTL module *1

1.2 Experiment Wiring

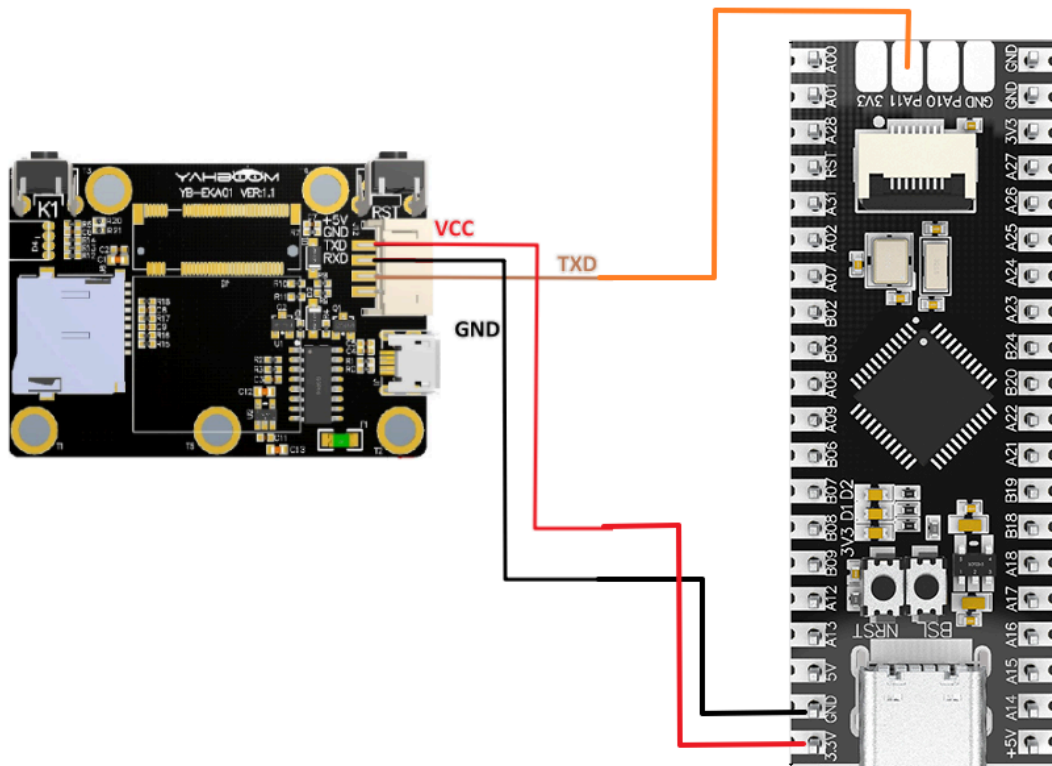
Method 1: Use USB to TTL module

MSPM0G	K210 Vision Module
RX	TXD
GND	GND
VCC	5V

MSPM0G	USB to TTL Module
TX	RXD
GND	GND



Method 2: Use onboard Type-C port



1.3 Main Code Explanation

This example sets the serial port print baud rate to 115200 bps, and the K210 module connection baud rate to 115200 bps.

The screenshot displays the TI Studio configuration interface for the UART peripheral. The left sidebar shows the project configuration tree with 'UART' selected under 'COMMUNICATIONS (6)'. The main panel shows the 'UART (2 of 4 Added)' configuration for 'MYUART'. The 'Basic Configuration' section is expanded, showing the 'UART Initialization Configuration' with a 'Target Baud Rate' of 115200.

Project Configuration Tree (Left Sidebar):

- PROJECT CONFIGURATION...
 - Project Config... 1/1 ✓ (+)
- MSPM0 DRIVER LIBRARY ...
- SYSTEM (9)
 - Board 1/1 ✓ (+)
 - DMA (+)
 - GPIO (+)
 - MATHACL (+)
 - Configuration NVM (+)
 - RTC (+)
 - SYSCTL 1/1 ✓ (+)
 - SYSTICK 1/1 ✓ (+)
 - WWDT (+)
- ANALOG (6)
 - ADC12 (+)
 - COMP (+)
 - DAC12 (+)
 - GPAMP (+)
 - OPA (+)
 - VREF (+)
- COMMUNICATIONS (6)
 - I2C (+)
 - I2C - SMBUS (+)
 - MCAN (+)
 - SPI (+)
 - UART 2/4 ✓ (+)
 - UART - LIN (+)
- TIMERS (6)
 - TIMER - CAPTURE (+)

UART (2 of 4 Added) Configuration (Main Panel):

Buttons: + ADD, REMOVE ALL

Selected UARTs: MYUART, K210_Uart

Configuration Fields:

- Name: MYUART
- Selected Peripheral: UART0
- Quick Profiles: ^
- UART Profiles: Custom
- Basic Configuration: ^
- UART Initialization Configuration: ^

UART Initialization Configuration Fields:

- Clock Source: BUSCLK
- Clock Divider: Divide by 1
- Calculated Clock Source: 32.00 MHz
- Target Baud Rate: 115200
- Calculated Baud Rate: 115211.52
- Calculated Error (%): 0.01
- Word Length: 8 bits
- Parity: None
- Stop Bits: One
- HW Flow Control: Disable HW flow control

Type Filter Text...
X
<<
<
>
Software > UART

PROJECT CONFIGURATION...

Project Config... 1/1 ✓ (+)

MSPM0 DRIVER LIBRARY ...

SYSTEM (9)

Board 1/1 ✓ (+)

DMA (+)

GPIO (+)

MATHACL (+)

Configuration NVM (+)

RTC (+)

SYSCTL 1/1 ✓ (+)

SYSTICK 1/1 ✓ (+)

WWDT (+)

ANALOG (6)

ADC12 (+)

COMP (+)

DAC12 (+)

GPAMP (+)

OPA (+)

VREF (+)

COMMUNICATIONS (6)

I2C (+)

I2C - SMBUS (+)

MCAN (+)

SPI (+)

UART 2/4 ✓ (+)

UART - LIN (+)

TIMERS (6)

TIMER - CAPTURE (+)

TIMER - COMPARE (+)

UART (2 of 4 Added) ②

✓ MYUART

✓ K210_Uart

Name

K210_Uart

Selected Peripheral

UART1

Quick Profiles

UART Profiles

Custom

Basic Configuration

UART Initialization Configuration

Clock Source

BUSCLK

Clock Divider

Divide by 1

Calculated Clock Source

32.00 MHz

Target Baud Rate

115200

Calculated Baud Rate

115211.52

Calculated Error (%)

0.01

Word Length

8 bits

Parity

None

Stop Bits

One

HW Flow Control

Disable HW flow control

```

int main(void)
{
    SYSCFG_DL_init();

    NVIC_ClearPendingIRQ(MYUART_INST_INT_IRQN); // Clear serial interrupt flag
    NVIC_EnableIRQ(MYUART_INST_INT_IRQN); // Enable serial interrupt

    while (1)
    {
        if (k210_msg.class_n != 0) // If class number is not empty
        {
            if(k210_msg.class_n == 1) // If color recognition
            {
                sprintf(buff_com,"class =
%d,x=%d,y=%d,w=%d,h=%d\r\n",k210_msg.class_n,k210_msg.x,k210_msg.y,k210_msg.w,k210_msg.h);
                uart0_send_string(buff_com);
                k210_msg.class_n = 0;
            }
        }
    }
}

```

```

    }

    }
    delay_ms(500);

}
}

```

Through the above program, if running this example, the k210_msg structure members will have corresponding values and be printed through serial port.

k210_msg: is the structure that receives information. Its main members are:

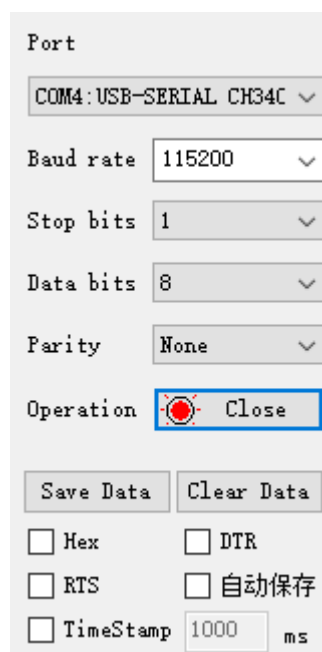
- x: is the x-coordinate of the upper-left corner of the recognized box (range: 0-240)
- y: is the y-coordinate of the upper-left corner of the recognized box (range: 0-320)
- w: is the width of the recognized box (range: 0-240)
- h: is the height of the recognized box (range: 0-320)
- id: is the recognized label
- class_n: class number
- msg_msg[20]: valid data

After data reception function and processing, each member of k210_msg will store valid information. If you want secondary development, you can directly call k210_msg members.

Note: Keil project source code must be placed under SDK path for compilation.

1.4 Experimental Phenomenon

1. After connecting the wires, run the K210 vision module offline [0. K210 Prerequisites For Making A Coprocessor - Must-see]
2. Set the serial assistant interface as shown in the figure.



3. Then run the color recognition example. The serial assistant will print important information

transmitted by K210 to MSPM0G, as shown in the figure.

```
x=0, y=0, w=319, h=106
x=0, y=0, w=319, h=95
x=0, y=0, w=319, h=85
x=0, y=0, w=319, h=73
x=177, y=0, w=142, h=62
x=192, y=0, w=127, h=55
x=197, y=0, w=122, h=52
x=202, y=0, w=117, h=50
x=204, y=0, w=115, h=48
x=201, y=0, w=118, h=49
x=202, y=0, w=117, h=49
x=206, y=0, w=113, h=47
x=209, y=0, w=110, h=47
x=209, y=0, w=110, h=47
x=209, y=0, w=110, h=47
x=211, y=0, w=108, h=46
x=215, y=0, w=104, h=44
x=216, y=0, w=103, h=44
x=216, y=0, w=103, h=43
x=218, y=0, w=101, h=43
x=219, y=0, w=100, h=42
x=219, y=0, w=100, h=42
x=218, y=0, w=101, h=41
x=219, y=0, w=100, h=40
x=224, y=0, w=95, h=36
```